Part 1 : SQL Fundamentals review

1 What is SQL ?
> SQL is a standard programming language used to communicate with and manage data in relational databases.
2 Why is SQL important in database systems?
> its ability to provide a bridge between users and complex data storage, allowing efficient management, analysis, and protection of information.
3. Explain the difference between:
.DDL
.DML
.DCL
.TCL
> DDL (data definition language) : purpose define or change structure of the database like altering and deleting database structure such as tables, indexes and schemas.

>DML (Data manipulation language) : are used to manipulate data stored in database tables. With BML you can insert new record update, delete and retrieve.

> DQL (Data query language) : used to fetch data from the database.

> TCL (Transaction control language) : a set of tasks into a single execution unit. Each transaction begins with a specific task and ends when all the tasks in the group are successfully completed. If any task fail transaction fail

Part 2: Query Analysis

For each query below:
● Explain what the query does
● Describe a real-world use case for the query

Query A
SELECT * FROM employees;
>What it does : retrieve all the columns and all rows from employees table
> Real world use : quick inspection of a table during development or debugging.
Ex : An hr admin reviewing the full employee dataset
Query B
SELECT department, AVG(salary)
FROM employees
GROUP BY department;
> What it does : Group employee by department calculate the average salary per department and return one row per department

> Real world use case : Salary benchmarking across departments. Ex : Hr analysing whether Engineering pay more than marketing

Query C
```
SELECT c.name, o.total
FROM customers c
JOIN orders o ON c.customer_id = o.customer_id;
```
> What it does : Perform an inner join between customers and orders match rows where customer id is the same in both tables. Return Customer name and order total.
> Real world use : Generating sales report showing who bought what. Ex : e commerce system listing customer purchase.

Part 3: SQL in Real-World Applications

Choose three (3) industries and explain how SQL is used in each.
Possible industries include:
.Banking
.Healthcare
.E-commerce
.Education
.Government
.Logistics
etc.
For each industry, discuss:
1. What types of data are stored?
2. How SQL is used to manage/analyze the data
3. Example SQL-related tasks in that industry

1. Banking

> Type of data :
- Customer profiles
- Account data
- Transaction
- Loan record
- Audit log

> SQl is used :
- Transaction integrity : sql databases enforce ACID properties for money movement
- Reporting : daily balance regulatory report
- Concurrency control : Prevents race conditions
- Fraud detection queries : pattern analysis

> Example SQL tasks :
- Transfer money safely using transaction
- Detect suspicious activity

2. Healthcare

> Type of data :
- Patient records
- Medical records
- Lab results
- Appointment
- Billing

> How SQL is used :
- Patient record retrieval
- Clinical analysis
- Scheduling system
- Data integrity

> Example SQL tasks :
- Retrieve full patient history
- Find the specific patient condition
- bill history

3. E commerce

    > Type of data :
    - Customers
    - Product
    - Order
    - Payments

    > How SQL is used :
- Order processing
- Inventory management
- Customer analytics

    > Example SQL tasks :
- Track sales per product
- Identify top customer/ product

Part 4: Reflection

1. Which SQL concept do you find most challenging? Why?
> Sql concept most challenging is join and query because
Must know about relationship between tables not single dataset

2. Which SQL concept do you believe is most important for industry use? Why?
> The most important concept for industry use would be SELECT statements or Transaction and data integrity because in the real system, correctness is more valuable than convenience and industry depends on it like healthcare banking and logistics.

3. How does SQL support data-driven decision making?

> SQL support data driven decision making by by transforming raw data into structure, queryable insights through

- Data retrieval
- Aggregation and summarization
- Pattern identification
- Decision support